

Basic Definitions

Differential Equation:

An equation containing the derivatives or differentials of one or more dependent variables with respect to one or more independent variables is said to be a differential equation (DE).

Example: $\frac{dy}{dt} + y = 3e^t, xdy + ydx = 0$.

Differential equations are classified according to type, order and linearity.

Classification according to type:

1. Ordinary Differential equation (ODE):

If an equations contains only ordinary derivatives of one or more dependent variables with respect to a single independent variable, it is then said to be an ordinary differential equation (ODE).

Example: $\frac{dy}{dt} - 5y = 0, (y - x)dx + 4xdy = 0, \frac{du}{dx} - \frac{dv}{dx} = x, \frac{d^2y}{dx^2} - 2\frac{dy}{dx} + 6y = 0$.

2. Partial Differential Equation (PDE):

An equation involving the partial derivatives of one or more dependent variables of two or more independent variables is called a partial differential equation (PDE).

Example: $\frac{\partial u}{\partial y} = \frac{\partial v}{\partial x}, x\frac{\partial u}{\partial x} + y\frac{\partial u}{\partial y} = u, \frac{\partial^2 u}{\partial x^2} = \frac{\partial^2 u}{\partial t^2} - 2\frac{\partial u}{\partial t}$.

Classification of order:

The order of the highest-order derivative in a differential equation is called the order of the equation.

Example: The second order ordinary differential equation is:

$$\frac{d^2y}{dx^2} + 5\left(\frac{dy}{dx}\right)^3 - 4y = e^x.$$

The first order ordinary differential equation is:

$$4t\frac{dy}{dt} + y = t.$$

The fourth order partial differential equation is:

$$a^2\frac{\partial^4 u}{\partial x^4} + \frac{\partial^2 u}{\partial t^2} = 0.$$

A general nth order ordinary differential equation is:

$$F\left(x, y, \frac{dy}{dx}, \frac{d^2y}{dx^2}, \dots, \frac{d^ny}{dx^n}\right) = 0.$$

Classification according to linearity:

1. Linear Equation:

A differential equation is said to be linear if it can be written in the form:

$$a_n(x) \frac{d^ny}{dx^n} + a_{n-1}(x) \frac{d^{n-1}y}{dx^{n-1}} + \dots + a_1(x) \frac{dy}{dx} + a_0(x)y = g(x).$$

$$\text{Example: } xdy + ydx = 0, \quad y-2y' + y'' = 0, \quad x^3 \frac{d^3y}{dx^3} - x^2 \frac{d^2y}{dx^2} + 3x \frac{dy}{dx} + 5y = e^x$$

are linear first, second and third order ODE.

Two properties:

The linear differential equations are characterized by two properties:

- (i) The dependent variable y and all its derivatives are of the first degree that is power of each term involving y is 1.
- (ii) Each coefficient depends on only the independent variable x .

2. Non-linear Equation:

An equation that is not linear is said to be nonlinear.

$$\text{Example: } yy'' - 2y' = x; \text{ [coefficient depends on } y]$$

$$\text{and } \frac{d^3y}{dx^3} + y^2 = 0; \text{ [the power of } y \text{ is not 1].}$$

Above are non-linear second order and third order ODEs.

Classification of Linear DE by Coefficient:

- Differential equation with constant-coefficient,
- Differential equation with variable-coefficient.

Solution of a differential equation:

A solution of a differential equation is an expression for the dependent variable in terms of the independent one(s) which satisfies the relation. The general solution includes all possible solutions and typically includes arbitrary constants (in the case of an ODE) or arbitrary functions (in the case of a PDE).

1. Trivial solution:

A solution of a differential equation that is identically zero on an interval I is often referred to as a trivial solution.

Example: $y = f(x) = 0$, $y'' + y' + y = 0$.

2. Explicit solution:

A solution of an ODE that can be written in the form $y = f(x)$ is called explicit solution.

Example: $y = e^{x^2}$ is an explicit solution of $\frac{dy}{dx} = 2xy$.

3. Implicit solution:

A relation $G(x, y) = 0$ is called an implicit solution of an ODE on an interval I provided it defines one or more explicit solutions on I .

Example: $f(x, y) = x^2 + y^2 - 4 = 0$ is an implicit solution of $\frac{dy}{dx} = -\frac{x}{y}$.

4. Parameter solution:

When solving an n th order equation $F(x, y, y', \dots, y^{(n)}) = 0$ where $y^{(n)}$ means $d^n y/dx^n$, we expect an n -parameter family of solutions, $G(x, y, c, \dots, c_n) = 0$. If a solution consists with one constant term is called one parameter solution.

5. Particular solution:

A solution of a DE that is free of arbitrary parameters is called a particular solution.

Example: $y = ce^x$ is a one parameter family of solutions of the simple first order equation $y = y'$. For $c = 2$ we get the particular solution $y = 2e^x$.

6. Singular solution:

A solution of a differential equation that cannot be obtained by specializing the parameters in a family of solutions is called singular solution.

Example: One parameter family of solutions of $y' = xy^{1/2}$ is given by $y = \left(\frac{x^2}{4} + c\right)^2$. When $c = 0$, the resulting particular solution is $y = x^4/16$. In this case the trivial solution $y = 0$ is singular solution of the equation, since it cannot be obtained from the family for any choice of the parameter.

7. General solution:

If every solution of $f(x, y, y', \dots, y^{(n)}) = 0$ on an interval I can be obtained from $G(x, y, c, \dots, c_n) = 0$ by appropriate choices of the c_i , $i = 1, 2, \dots, n$. We then say that the n -parameter family is the general or complete solution of the differential equation.

Degree of a differential equation:

The degree of an algebraic differential equation is the degree of the derivative of the highest order in the equation after the equation is freed from radicals and fraction in its derivatives.

Example: $\frac{dy}{dx} = \sqrt{1 + \frac{d^2y}{dx^2}}$; i. e. $\left(\frac{dy}{dx}\right)^2 = 1 + \frac{d^2y}{dx^2}$ is a 2nd order and 1st degree differential equation.

Geometrical Interpretation:

A differential equation represents a family of curves all satisfying some common properties.

Separable equation:

A differential equation of the form $\frac{dy}{dx} = \frac{g(x)}{h(y)}$ is said to be separable or to have separable variables.

Homogeneous function:

If a function has the property that $f(tx, ty) = t^n f(x, y)$ for some real number n , then f is called a homogeneous function of degree n .

Homogeneous equation:

A differential equation of the form $M(x, y)dx + N(x, y)dy = 0$ is called homogeneous if both coefficients M and N are homogeneous functions of the same degree.

Exact equation:

A differential expression $M(x, y) dx + N(x, y) dy$ is an exact differential in a region R of the xy -plane if it corresponds to the total differential of some function $f(x, y)$. A differential equation of the form $M(x, y) dx + N(x, y) dy = 0$ is called an exact equation if the expression on the left side is an exact differential.

Criterion for an exact differential:

Let $M(x, y)$ and $N(x, y)$ be continuous and have continuous first partial derivatives in a rectangular region R defined by $a < x < b$, $c < y < d$. Then a necessary and sufficient condition is that $M(x, y) dx + N(x, y) dy$ be an exact differential is $\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}$.

Proof of the necessity:

For simplicity let us assume that $M(x, y)$ and $N(x, y)$ have continuous first partial derivatives for all (x, y) . Now if the expression $M(x, y) dx + N(x, y) dy$ is exact, there exist any function f for which $M(x, y) dx + N(x, y) dy = \frac{\partial f}{\partial x} dx + \frac{\partial f}{\partial y} dy$ for all (x, y) in R .

Therefore, $M(x, y) = \frac{\partial f}{\partial x}$, $N(x, y) = \frac{\partial f}{\partial y}$.

And $\frac{\partial M}{\partial y} = \frac{\partial}{\partial y} \left(\frac{\partial f}{\partial x} \right) = \frac{\partial^2 f}{\partial y \partial x} = \frac{\partial}{\partial x} \left(\frac{\partial f}{\partial y} \right) = \frac{\partial N}{\partial x}$.

The equality of the mixed partials is a consequence of the continuity of the first partial derivatives of $M(x, y)$ and $N(x, y)$.

Integrating Factor:

When a non-exact differential equation is possible to convert into an exact differential equation by multiplying a function $\mu(x, y)$ called an integrating factor.

However the resulting exact equation $\mu M(x, y)dx + \mu N(x, y)dy = 0$ may not be equivalent to the original equation in the sense that a solution of one is also a solution of the other. It is possible for a solution to be lost or gained as a result of the multiplication.

Bernoulli's Equation:

The differential equation:

$$\frac{dy}{dx} + p(x)y = f(x)y^n$$

where n is any real number is called Bernoulli's equation. We observe that, if $n = 0$, the equation reduces to a first order linear differential equation. And if $n = 1$, then the linear equation is a separable equation.

Riccati's Equation:

The non-linear differential equation

$$\frac{dy}{dx} = P(x) + Q(x)y + R(x)y^2$$

is called Riccati's equation.

Clairaut's Equation:

A solution of Clairaut's equation

$$y = xy' + f(y')$$

is the family of straight lines $y = cx + f(c)$, where c is an arbitrary constant.

Orthogonal Curve:

Two curves l_1 and l_2 are said to be orthogonal at a point if and only if their tangent lines T_1 and T_2 are perpendicular, which means that $m_1 m_2 = -1$ at the point of intersection. For example, the graphs of $y = (-1/2)x + 1$ and $y = 2x + 4$ are perpendicular.

Orthogonal Trajectories:

When all the curves of one family of curves $G(x, y, c_1) = 0$ intersect orthogonally all the curves of another family $H(x, y, c_2) = 0$, then the families are called orthogonal trajectories of each other.

In other words, an orthogonal trajectory is any one curve that intersects every curve of another family of right angles.

Example: The graph of $y = (-1/2)x + 1$ is an orthogonal trajectory of $y = 2x + c_1$. The families $y = (-1/2)x + c_2$ and $y = 2x + c_1$ are orthogonal trajectories.

Orthogonal trajectories occur naturally in the constructions of meteorological maps and in the study of electricity of magnetism.

Differential equation with variable coefficients:

Cauchy-Euler Equation

Any linear differential equation of the form

$$a_n(x) \frac{d^n y}{dx^n} + a_{n-1}(x) \frac{d^{n-1} y}{dx^{n-1}} + \cdots + a_1(x) \frac{dy}{dx} + a_0(x)y = g(x)$$

where $a_n, a_{n-1}, \dots, a_0$ are constants and the equation is called Cauchy-Euler equation or equidimensional equation.

This type of differential equation is that the degree of each monomial coefficient matches the order of differentiation.

Initial Value Problem (IVP):

We are to obtain by solving 1st order DE $\frac{dy}{dx} = f(x, y)$ subject to the side condition $y(x_0) = y_0$ where x_0 is a number in an interval I and y_0 is any arbitrary real number.

The problem: Solve $\frac{dy}{dx} = f(x, y)$

Subject to: $y(x_0) = y_0$ is called an initial value problem and the side condition is known as the initial condition.

Initial value problem for a linear n-th order DE is

$$\text{Solve: } a_n(x) \frac{d^n y}{dx^n} + a_{n-1}(x) \frac{d^{n-1} y}{dx^{n-1}} + \cdots + a_1(x) \frac{dy}{dx} + a_0(x)y = g(x)$$

Subject to $y(x_0) = y_0, y'(x_0) = y'_0, \dots, y^{(n-1)}(x_0) = y_0^{(n-1)}$; where $y_0, y'_0, \dots, y_0^{(n-1)}$ are arbitrary constants, is called an initial value problem.

The specified values $y(x_0) = y_0, y'(x_0) = y'_0, \dots, y^{(n-1)}(x_0) = y_0^{(n-1)}$ are initial conditions. We seek a solution on some interval I containing x_0 .

Boundary Value Problem (BVP):

For solving a differential equation of order two or greater in which the dependent variable y or its derivatives are specified at different points. A problem such as

$$a_2(x) \frac{d^2 y}{dx^2} + a_1(x) \frac{dy}{dx} + a_0(x)y = g(x)$$

Subject to: $y(a) = y_0$, $y(b) = y_1$ is called a two-point boundary value problem or simply boundary value problem.

The specified values $y(a) = y_0$, $y(b) = y_1$ are called boundary conditions.

Eigenvalues and Eigenvectors:

Let A be an $n \times n$ matrix. A number λ is called an eigenvalue of A if there exists a nonzero solution vector v of the linear system $Av = \lambda v$. The solution vector v is called an eigenvector corresponding to the eigenvalue λ .

Solution vector:

A solution vector on an interval I is any column matrix $X = \begin{pmatrix} x_1(t) \\ x_2(t) \\ \vdots \\ x_n(t) \end{pmatrix}$, whose entries are differentiable functions satisfying the system $\frac{dX}{dt} = A(t)X + F(t)$ on the interval.

Fundamental Matrix:

Let $X_1 = \begin{pmatrix} x_{11} \\ x_{21} \\ \vdots \\ x_{n1} \end{pmatrix}$, $X_2 = \begin{pmatrix} x_{12} \\ x_{22} \\ \vdots \\ x_{n2} \end{pmatrix}$, ..., $X_n = \begin{pmatrix} x_{1n} \\ x_{2n} \\ \vdots \\ x_{nn} \end{pmatrix}$ be a fundamental set of n solution vectors of the homogeneous system $\frac{dX}{dt} = A(t)X$ on an interval I . The matrix

$$X(t) = \begin{pmatrix} x_{11} & x_{12} & \cdots & x_{1n} \\ x_{21} & x_{22} & \cdots & x_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ x_{n1} & x_{n2} & \cdots & x_{nn} \end{pmatrix}$$

is called a fundamental matrix of the system on the interval.